Question #1 of 74

Question ID: 1675256

Oskar Dieter manages a fund with exposures to both German equity and fixed income. The fund has a market value of €450m, with 60% exposure to equity and 40% to fixed income. The equity portfolio has a beta of 1.4 relative to the DAX 30, and the fixed-income portfolio has a modified duration of 12.6.

Dieter wishes to adjust the asset allocation to 70% equity and 30% fixed income using the following futures contracts:

DAX 30 Futures		Euro Bund Futures	
Futures price	13,400	Futures price	€174.50
		Contract size	€100,000
		CTD price	€99.90
Multiplier	€25	Conversion factor	0.5723
		CTD modified duration	9.86

Taking which of the following positions is *most likely* to achieve Dieter's required asset allocation?

A) Long 188 Dax 30 Index futures, short 329 Euro Bund futures.

B) Long 329 Euro Bund futures, short 134 Dax 30 Index futures.

C) Long 134 Dax 30 Index futures, short 576 Euro Bund futures.

Explanation

Stock	Current	Target	Change in Exposure
Equity	€270m (60%)	€315m (70%)	+€45m
Fixed income	€180m (40%)	€135m (30%)	−€45m
Total	€450m	€450m	

DAX 30 Futures

$$\text{Number of futures required} = \left(\frac{\beta_T - \beta_P}{\beta_F}\right)\left(\frac{MV_P}{F}\right)$$

where:

β_T = target portfolio beta

β_P = current portfolio beta

β_F = futures beta (beta of stock index)

MV_P = market value of portfolio

F = futures contract value = futures price × multiplier

$$\text{Number of futures required} = \left(\frac{1.4-0}{1}\right)\left(\frac{€45{,}000{,}000}{13{,}400 \times €25}\right) = +188.06$$

Dieter will need to buy 188 futures contracts to achieve his target asset allocation.

Euro Bund Futures

$$\text{BPV HR} = \frac{\text{BPV}_{\text{Target}} - \text{BPV}_{\text{Portfolio}}}{\text{BPV}_{\text{CTD}}} \times \text{CF}$$

$BPV_{Portfolio} = MD_{Portfolio} \times 0.0001 \times MV_{Portfolio} = 12.6 \times 0.0001 \times €45{,}000{,}000 = €56{,}700$

$BPV_{CTD} = MD_{CTD} \times 0.0001 \times [\text{Price} / 100 \times \text{contract size}] = 9.86 \times 0.0001 \times [(€99.90 / 100) \times €100{,}000] = €98.50$

$$\text{BPV HR} = \frac{€0 - €56{,}700}{€98.50} \times 0.5723 = -329.44$$

Dieter will need to sell 329.44 German Bund future contracts.

(Module 17.5, LOS 17.f)

Question #2 of 74

Question ID: 1675249

Tom Corser is the manager of the $140,000,000 Intrepid Growth Fund. Corser's long-term view of the equity market is negative, and as a result, his portfolio is allocated defensively with a beta of 0.85. Despite his negative long-term outlook, Corser thinks the market is temporarily mispriced and could rise significantly over the next few weeks. Corser has implemented tactical asset allocation measures in his fund sporadically over the years, and he thinks now is another time to do so. Because he likes his long-term holdings, he decides to use a futures overlay rather than trading assets to implement his view of the market. Corser decides he wants to increase the beta of his portfolio to 1.25. The appropriate futures contract has a beta of 1.03, and the total futures price is $310,000. What is the appropriate tactical allocation strategy for Corser to accomplish his objective?

A) Buy 373 equity futures contracts.

B) Sell 175 equity futures contracts.

C) Buy 175 equity futures contracts.

Explanation

Note: On the exam, it is very likely for material on tactical asset allocation to be tested in conjunction with material from derivatives as tactical asset allocation can be accomplished by selling assets, or with a derivative overlay. Because Corser wants to increase the beta of his portfolio, he should buy futures contracts. The appropriate number of contracts to buy is calculated as follows:

$$[(1.25 - 0.85) / 1.03] \times ($140,000,000 / $310,000) = 175.38 \approx 175 \text{ contracts.}$$

(Module 17.5, LOS 17.f)

Question #3 of 74

Question ID: 1675252

A manager of $30 million in mid-cap equities would like to move half of the position to an exposure resembling small-cap equities. The beta of the mid-cap position is 1.0, and the average beta of small-cap stocks is 1.6. The betas of the corresponding mid- and small-cap futures contracts are 1.05 and 1.5, respectively. The mid- and small-cap futures prices are $260,000 and $222,222, respectively. What is the appropriate strategy?

A) Short 17 mid-cap futures and go long 17 small-cap futures.

B) Short 17 small-cap futures and go long 17 mid-cap futures.

C) Short 55 mid-cap futures and go long 72 small-cap futures.

Explanation

We should recall our formula for altering beta:

number of contracts = ({target beta − $B_{portfolio}$} × V) / ($B_{futures}$ × futures price)

In this case, for the first step where we convert the mid-cap position to cash, V = $15 million, and the target beta is 0. The current beta is 1.0, and the futures beta is 1.05:

–54.95 = (0 − 1) × ($15,000,000) / (1.05 × $260,000)

The manager should short 55 of the futures on the mid-cap index. Then, the manager should take a long position in the following number of contracts on the small-cap index:

72.00 = (1.6 − 0) × ($15,000,000) / (1.5 × $222,222)

Thus, the manager should take a long position in 72 of the contracts on the small-cap index.

(Module 17.5, LOS 17.f)

Question #4 of 74

Question ID: 1675204

Matthew Ricky runs a charity and is expecting to receive a donation of $20,000,000 in six months' time. He wishes to temporarily hold the money in a deposit account for three months, earning $LIBOR + 50 basis points, while he decides how best to use the funds. Ricky is worried that interest rates may decline before the deposit is made and decides to hedge the risk using Eurodollar futures. The six-month Eurodollar future is quoted at 97.50, with a contract size of $1,000,000. Six months later, Ricky receives the donation and makes the deposit when three-month USD LIBOR is quoted at 2%. Ricky closes out his futures position at a price of 98. The return that Ricky earned on his deposit over the three-month period is *closest* to:

A) $125,000.

B) $75,000.

C) $150,000.

Explanation

Ricky will need to purchase Eurodollar futures if he wishes to lock into a guaranteed interest rate for his deposit.

There is a key point with Eurodollar futures:

Selling futures = notional borrowing

Buying futures = notional lending

Ricky will require $20,000,000 / $1,000,000 (i.e., purchase 20 contracts).

Interest rate locked in = 100 − 97.50 = 2.5%

Margin above LIBOR on loan = 50 basis points (0.005)

This is the return over three-month deposit period:

$20,000,000 × (0.025 + 0.005) × 90 / 360 = $150,000

(Module 17.1, LOS 17.a)

Question #5 of 74

Question ID: 1675247

Sheila manages a $100 million fixed-income portfolio. The portfolio duration is currently 4.9, and she would like to increase it to 5.7. She selects a swap with a net duration of 6.1. Based only on the information provided, which of the following statements is *least likely* correct?

A) Sheila should have a pay-floating/receive-fixed position in the swap.

B) Sheila should select a swap with a notional principal of approximately $13 million to achieve the desired duration.

C) Sheila should have a receive-floating/pay-fixed position in the swap.

Explanation

Because Sheila wants to increase the duration of the portfolio, she should have a pay-floating/receive-fixed position in the swap with a notional principal of $13,114,754, as computed here:

Notional principal = $100,000,000 × [(5.7 – 4.9) / 6.1] = $13,114,754

(Module 17.5, LOS 17.e)

Question #6 of 74

Question ID: 1675222

Shota Hideaki manages a Japanese equity fund with a market value of ¥21,712,000,000 and a beta relative to the Nikkei 225 of 0.75. Due to recent sales, a cash balance of ¥1,065,500,000 has built up, and Hideaki has become concerned about the cash drag. Hideaki wishes to create a cash overlay to equitize these funds. Hideaki is considering using Nikkei 225 futures and wishes the cash overlay to have the same beta as his existing portfolio. The current futures price is 21,500 with a multiplier of ¥1,000. To correctly execute the cash overlay, Hideaki would *most appropriately* purchase:

A) 44 contracts.

B) 50 contracts.

C) 37 contracts.

Explanation

$$\text{Number of futures required} = \left(\frac{\beta_T - \beta_P}{\beta_F}\right)\left(\frac{MV_P}{F}\right)$$

β_P = current portfolio beta of cash position = 0 (note that cash has a beta of 0)

MV_P = cash position

$$\text{Number of futures required} = \left(\frac{\beta_T}{\beta_F}\right)\left(\frac{MV_P}{F}\right)$$

β_T = target portfolio beta. In this case, the target will be set to 0.75 because we want the beta of the synthetically invested cash to match the beta of the existing portfolio.

$$\text{Number of futures required} = \left(\frac{0.75}{1}\right)\left(\frac{¥1{,}065{,}500{,}000}{21{,}500 \times ¥1{,}000}\right) = 37.17$$

$$\approx 37 \text{ futures contracts}$$

(Module 17.3, LOS 17.c)

Question #7 of 74

Question ID: 1675255

Nicholas Avery manages a U.S. equity fund with a market value of $990 million. Currently, the fund is invested 70% in large-cap equity and 30% in small-cap equity, but Avery wishes to alter the allocation to 60% in large-cap and 40% in small-cap equity. Avery's large-cap portfolio has a beta of 1.2 relative to the S&P 500, and his small-cap portfolio has a beta of 0.9 relative to the MSCI USA Small Cap Index. Avery plans to make use of the following futures contracts to achieve his target asset allocation:

	S&P 500 Futures	MSCI USA Small Cap Index Futures
Futures price	3,100	720
Index multiplier	$250	$100

Taking which of the following positions is *most likely* to achieve Avery's target asset allocation?

A) Buy 1,650 MSCI USA small-cap futures; sell 115 S&P 500 futures. ✗

B) Buy 1,375 MSCI USA small-cap futures; sell 128 S&P 500 futures. ✗

C) Buy 1,238 MSCI USA small-cap futures; sell 153 S&P 500 futures. ✓

Explanation

Stock	Current	Target	Change in Exposure
Large cap	$693m (70%)	$594m (60%)	−$99m
Small cap	$297m (30%)	$396m (40%)	+$99m
Total	$990m	$990m	

S&P 500 Futures

$$\text{Number of futures required} = \left(\frac{\beta_T - \beta_P}{\beta_F}\right)\left(\frac{MV_P}{F}\right)$$

where:

β_T = target portfolio beta

β_P = current portfolio beta

β_F = futures beta (beta of stock index)

MV_P = market value of portfolio

F = futures contract value = futures price × multiplier

Number of futures required = (0 − 1.2)[$99,000,000 / (3,100 × $250)] = −153.29

Avery will need to sell 153 futures contracts to achieve his target asset allocation.

MSCI USA Small Cap Index Futures

Number of futures required = (0.9 − 0)[$99,000,000 / (720 × $100)] = +1,237.50

Avery will need to buy 1,238 futures contracts to achieve his target asset allocation.

(Module 17.5, LOS 17.f)

Question #8 of 74

Question ID: 1675251

A manager of $40 million of mid-cap equities would like to move $5 million of the position to large-cap equities. The beta of the mid-cap position is 1.1, and the average beta of large-cap stocks is 0.9. The betas of the corresponding mid- and large-cap futures contracts are 1.1 and 0.95, respectively. The mid- and large-cap futures prices are $252,000 and $98,222, respectively. What is the appropriate strategy?

A) Short 29 mid-cap futures and go long 29 large-cap futures. ✗

B) Short 20 mid-cap futures and go long 48 large-cap futures. ✓

C) Short 23 mid-cap futures and go long 42 large-cap futures. ✗

Explanation

We should recall our formula for altering beta:

$$\text{number of contracts} = (\{\text{target beta} - B_{portfolio}\} \times V) / (B_{futures} \times \text{futures price})$$

In this case, for the first step where we convert the mid-cap position to cash, V = $5 million, and the target beta is 0. The current beta is 1.1, and the futures beta is 1.1:

$$-19.84 = (0 - 1.1) \times (\$5{,}000{,}000) / (1.1 \times \$252{,}000)$$

The manager should short 20 of the futures on the mid-cap index. Then, the manager should take a long position in the following number of contracts on the large-cap index:

$$48.23 = (0.9 - 0) \times (\$5{,}000{,}000) / (0.95 \times \$98{,}222)$$

Thus, the manager should take a long position in 48 of the contracts on the large-cap index.

(Module 17.5, LOS 17.f)

Question #9 of 74

Question ID: 1675211

If a manager shorts a forward currency contract to hedge the expected value of a foreign equity portfolio in one year, the worst-case scenario is if the portfolio's return is:

A) less than the expected value and the currency depreciates. ✗

B) less than the expected value and the currency appreciates. ✓

C) greater than the expected value and the currency appreciates. ✗

Explanation

This should be obvious because a decline in the equity position is bad, and the short position in a forward currency contract hurts when the foreign currency appreciates. If the equity position falls short of the contracted amount, in addition to the loss from the decline in asset prices, then the manager will suffer a loss equal to the difference in the hedged amount and the actual equity value times the difference in the spot and contracted forward rate.

(Module 17.2, LOS 17.b)

Question #10 of 74

Question ID: 1675246

A manager of a $40 million fixed-income portfolio with a duration of 4.2 wants to lower the duration to 3. The manager chooses a swap with a net duration of 2.1. What notional principal (NP) should the manager choose for the swap to achieve the target duration?

A) $56,000,000. ✗

B) $22,857,143. ✓

C) $70,000,000. ✗

Explanation

NP = $40,000,000 × (3 − 4.2) / –2.1

NP = $22,857,143

Because the manager wants to *reduce* the duration of his portfolio, he should take a receive-floating/pay-fixed position in the swap with that notional principal. Remember that a receive-floating swap has a negative duration, so we enter –2.1 in the equation.

(Module 17.5, LOS 17.e)

Question #11 of 74

Question ID: 1675213

CorePro Incorporated, based in the United States, wishes to borrow €30m to finance a subsidiary in Italy. CorePro has decided to borrow domestically in dollars and then enter into a cross-currency basis swap to exchange the USD for euros. The basis on the euro-dollar swap is quoted as −20 basis points. The current $/€ exchange rate is $1.2. The swap has a 10-year tenor with quarterly settlement dates. At the last settlement date, 90-day USD LIBOR was 2%, and 90-day Euribor was 2.5%. The interest paid and received by CorePro at the next settlement date will be *closest* to:

A) Pays $180,000 and receives €172,500. ✗

B) Pays €187,500 and receives $198,000. ✗

C) Pays €172,500 and receives $180,000. ✓

Explanation

A cross-currency swap pays variable interest on both legs, typically based on LIBOR or Euribor. When the dollar is trading at a positive basis against most currencies, it means the floating $ interest rate in a cross-currency swap will be higher than the $ LIBOR rate. Arbitrage should prevent this from happening but due to the additional regulatory cost (in particular implications for capital adequacy) fewer banks are undertaking these activities.

In this example the dollar has +20BP against the Euro. This means that dollar lenders will generate an additional 20 BP above $ LIBOR when lending the dollar. Rather than adding the 20BP to the $ interest received each period, typically the 20 BP is deducted from the Euro interest payments. An extra 20 BP on dollar lending will be quoted as a basis of −20BP against the Euro.

Next settlement date:

USD 90-day LIBOR at last settlement date = 2%

90-day Euribor at last settlement date = 2.5%

Basis quoted at initiation of the swap −20 BP (typically quoted from the non-dollar perspective)

CorePro (dollar lender, euro receiver):

Receives $36m × 0.02 × 90/360 = $180,000

Pays €30m × (0.025 − 0.002) × 90/360 = €172,500

Note that the dollar lender benefits from the 20 BP basis, not by receiving 20 BP more $ interest, but by paying 20 BP less € interest. Note that because the principal amounts are equivalent ($36m = €30m) at inception whether the basis is added to the $ interest payments or deducted from the € interest payments makes no difference. (However, market convention is to deduct it from the non-dollar interest payments.)

You would have arrived at an incorrect answer if you added the basis to the $ interest.

(Module 17.2, LOS 17.b)

Question #12 of 74

Question ID: 1675205

Which of the following comments relating to fixed-income (bond) futures is *least accurate*?

A) The seller (short) party has the option of which bond and when to deliver in the delivery month. ✗

B) The cheapest-to-deliver bond will have the highest basis and lowest repo rate. ✓

C) The futures price is calculated using the cheapest-to-deliver bond and a conversion factor. ✗

Explanation

The short party can deliver any eligible Treasury bond in the delivery month. The seller can choose when in the delivery month to deliver the bond.

Futures prices on sovereign debt are quoted for a notional bond. Each eligible bond is assigned a conversion factor (CF), computed by discounting the bond at a constant YTM, regardless of the bond's actual yield. Bonds with coupon rates lower than the YTM will have conversion factors of less than 1, and bonds with coupon rates greater than the YTM will have CFs greater than 1.

Before delivery, the quoted futures price is calculated by calculating the futures price based on the cheapest-to-deliver (CTD) bond and then dividing by the CTD conversion factor. This last step means that the price on the exchange is based on the notional bond rather than the CTD bond.

The cash and carry model is still used to price the future. The bond selected to price the future is the CTD. There are two methods to identify the CTD bond. The first method identifies the eligible bond that generates the highest return (implied repo rate) on a cash and carry trade. The second method used to identify the CTD bond is to find the eligible bond with the lowest basis. Basis is defined as the spot price − futures price.

The Treasury bond with the lowest basis will typically have the highest implied repo rate and be the cheapest to deliver.

(Module 17.1, LOS 17.a)

Question #13 - 16 of 74

Question ID: 1675264

Based on **Exhibit 1: Selected Information Pertaining to the Equity Index Futures Contracts (In EUR)**, to adjust the exposure of the pension fund's equity portfolio, Bright should *most likely*:

A) short 333 DAX futures contracts and long 2,262 CAC 40 futures contracts.

B) short 11,760 DAX futures contracts and long 16,019 CAC 40 futures contracts.

C) short 137 DAX futures contracts and long 660 CAC 40 futures contracts.

Explanation

To decrease the exposure to Germans stocks by 1.225 billion × 10% = €112.5 million and increase the exposure to French stocks by 1.225 billion × 10% = €122.5 million, Bright would have to short futures contracts on the DAX and long futures contracts on the CAC 40.

The number of future contracts to be sold or bought is calculated as:

$$N_f = \left(\frac{\beta_T - \beta_S}{\beta_f}\right)\left(\frac{S}{F}\right)$$

where *N*f represents the number of equity futures contracts, β_T is the target beta of the portfolio, β_s is the current beta of the portfolio, β_f is the beta of the futures contract, *S* is the size of the portfolio, and *F* is the actual futures price, which is calculated as quoted futures price × futures contract multiplier.

To achieve the desired exposure to the German stocks, S = €122.5 million and β_T = 0 (because the exposure needs to be eliminated):

$$N_{f,DAX} = \left(\frac{0 - 0.85}{1}\right)\left(\frac{122{,}500{,}000}{12{,}500 \times 25}\right) = -333.20$$

Thus, Bright needs to sell 333 futures contracts on the DAX index.

To achieve the desired exposure to the French stocks, S = €122.5 million and β_T = 1.20.

of CAC 40 contracts = (1.2 − 0 / 1) × (EUR 122.5 million / 6,500 × 10) = 2,261.54 (i.e., 2,262)

Therefore, Bright needs to buy 2,262 futures contracts on the CAC 40 index.

$N_{f,DAX}$ is miscalculated as $\left(\frac{0.85 - 1.2}{1}\right)\left(\frac{122{,}500{,}000}{12{,}500 \times 25}\right) = -137$ and $N_{f,CAC}$ 40 is miscalculated as

$$\left(\frac{1.2 - 0.85}{1}\right)\left(\frac{122{,}500{,}000}{6{,}500 \times 10}\right) = 660$$

The futures multipliers are excluded, resulting in $N_{f,DAX} = \left(\frac{0 - 1.2}{1}\right)\left(\frac{122{,}500{,}000}{12{,}500}\right) =$ −11,760, and $N_{f,CAC\ 40}\left(\frac{0.85 - 0}{1}\right)\left(\frac{122{,}500{,}000}{6{,}500}\right)$ = 16,019.

(Module 17.5, LOS 17.f)

Question #14 - 16 of 74

Question ID: 1675265

The total change in the market value of the French bond exposure due to the increase in the French bond yield is *closest* to:

A) –€1,379,035. ✗

B) –€1,496,250. ✗

C) –€1,375,410. ✓

Explanation

To benefit from her expectations, Bright will have to decrease the duration of the French bonds by selling futures contracts.

The total change in the value of the French bond exposure includes the change in the value of the French bond portfolio and the change in the value of the futures contracts on the French bonds:

- Change in value of the French bonds = –change in interest rate (in basis points) × BPV = –5 × 299,250 = –€1,496,250
- Change in value of the short futures contracts = –change in interest rate (in basis points) × $\frac{\text{BPV CTD}}{\text{CF}}$ × (–number of futures contracts) = –5 × $\frac{98.5}{0.97}$ × (–238) = €120,840 (note that the number of futures contracts is preceded by a minus sign because the contracts are shorted)

The total change in the value of French bond exposure = –1,496,250 + 120,840 = –€1,375,410.

The –€1,496,250 answer option is incorrect as the change in the value of the futures contracts is disregarded.

The change in the value of the futures contracts is miscalculated by omitting the conversion factors. This gives –5 × 98.5 × (–238) = €117,215, resulting in total change in value of French exposure = –1,496,250 + 117,215 = –€1,379,035.

(Module 17.1, LOS 17.a)

Question #15 - 16 of 74

Question ID: 1675266

Bright's note pertaining to a variance swap is *most likely* incorrect with regard to:

A) the payoff of a variance swap. ✗

B) the price of a variance swap. ✓

C) the equivalence of a variance swap to a portfolio consisting of a long position in a basket of options and a short position in the underlying asset. ✗

Explanation

Variance swaps are instruments used by investors to take directional bets on implied versus realized volatility; these instruments are called *swaps* because they have payoffs analogous to the payoffs of swaps. In a variance swap, the contract buyer will pay the difference between the predetermined fixed variance strike and the realized annualized variance on the underlying over the specified period and applied to a variance notional.

Being long a variance swap is equivalent to being long a basket of options and short the underlying asset. As a result, the payoff of a variance swap is convex in volatility; this is attractive to (1) investors who desire a long volatility position to hedge tail risk and (2) volatility sellers who are able to sell the variance swap at a price that is *higher* than that of an at-the-money option.

The payoff of a variance swap is convex in volatility.

A variance swap is equivalent to a portfolio that consists of a long position in a basket of options and a short position in the underlying asset.

(Module 17.4, LOS 17.d)

Question #16 - 16 of 74

Question ID: 1675267

The profit on the variance swap purchased by Bright on the DAX index is *closest* to:

A) €95,000. ✓

B) €45,000. ✗

C) €2,500. ✗

Explanation

In a variance swap, the contract buyer will pay the difference between the predetermined fixed variance strike and the realized annualized variance on the underlying over the specified period and applied to a *variance notional*. However, variance swap traders typically express (1) trade sizes in *vega notional* and (2) the strike in volatility (not variance).

The vega notional represents the average profit and loss on a variance swap for a 1% increase in the volatility of the underlying. Thus, profit on the variance swap can be approximated as: 45,000 × 2 = €90,000.

The €45,000 answer option is incorrect as the figure represents the approximate change in the value of the variance swap for a 1% change in volatility.

The €2,500 answer option is incorrect as the figure is based on the variance notional, which is equal to vega notional / (2 × K) = 45,000 / (2 × 18) = 1,250 resulting in profit = 1,250 × 2 = €2,500.

The €95,000 answer option is correct and based on the variance notional and the squared difference between the realized volatility squared and strike price squared. Variance notional = vega notional / (2 × K) = 45,000 / (2 × 18) = 1,250, resulting in profit = 1,250 × ($20^2 – 18^2$) = €95,000.

(Module 17.4, LOS 17.d)

Question #17 of 74

Question ID: 1675225

Which of the following statements regarding variance swaps is *least accurate*?

A) The payoff on a variance swap is determined by the difference between actual volatility squared over a given period versus implied volatility squared. ✗

B) The payoff on a volatility swap is convex, resulting in gains that increase in relative magnitude as realized volatility increases. ✓

C) The purchaser of a variance swap will profit if the realized volatility is greater than implied volatility. ✗

Explanation

Volatility and variance swaps in practice are futures contracts rather than swaps. The payoff on a volatility swap is based on implied volatility versus actual (realized) volatility. The payoff on a variance swap is based on implied variance versus realized variance. Both swaps can be viewed as a bet on actual versus implied volatility.

The purchaser of both volatility and variance swap will benefit if realized variance or volatility is greater than the implied value. The implied volatility is derived from option prices.

$$\text{Payoff on a volatility swap} = (\sigma - K) \times \text{vega notional}$$

$$\text{Payoff on a variance swap} = (\sigma^2 - K^2) \times \text{variance notional}$$

where:

σ = realized volatility

K = implied volatility

$$\text{Variance notional} = \frac{\text{Vega notional}}{2 \times \text{Strike price (K)}}$$

The result of these payoff calculations is that a volatility swap generates a linear payoff, while a variance swap generates a convex payoff. The payoff on the variance swap is convex with volatility, which means that as realized volatility increases, the payoff increases at an accelerating rate.

Convexity is an attractive feature for the purchaser of the swap. For participants with equity exposures, purchasing the variance swap can help to offset tail risk. When volatility increases dramatically and equity values fall (the definition of tail risk), the decrease in the portfolio is offset by profits—increasing at an accelerating rate—on the variance swap.

(Module 17.4, LOS 17.d)

Question #18 of 74

Question ID: 1675210

A maker of large computers has just received an order for some of its products. The agreed-upon price is in British pounds: £8 million. The firm will receive the pounds in 60 days. The current exchange rate is $1.32/£, and the 60-day forward rate is $1.35/£. If the firm uses the forward contract to hedge the corresponding exchange rate risk, how many dollars will it expect to receive?

A) $10,560,000.00. ✗

B) $5,925,926.00. ✗

C) $10,800,000.00. ✓

Explanation

On the day the order comes in, the firm effectively has a long position in pounds; therefore, it should take a short position in a forward contract. This contract would obligate the firm to deliver the pounds that it will receive for dollars. The contract would be to exchange £8 million for the following:

$10,800,000 = (£8,000,000) × $1.35/£

(Module 17.2, LOS 17.b)

Question #19 of 74

Question ID: 1675221

Sofia Chiara manages an equity fund that invests in Italian stocks. The fund has a market value of €120m at the start of the period. Chiara sold 563 futures contracts at the start of the period to give her overall portfolio a beta (relative to the FTSE MIB) of 0.8. The equity portion of her portfolio has a beta of 1.3. The FTSE MIB has a value of 21,350 at the start of the period, and the futures price was 21,315.

Twelve months later, the index has fallen to 21,050, and the futures price is 21,025. Assuming that the beta remains constant at 1.3 over the period and the futures contract multiplier is €5 per index point, the return on her fund at the end of the 12-month period is *closest* to:

A) −€1,347,537.

B) −€1,348,946.

C) −€1,375,687.

Explanation

Equity portfolio performance:

% change in index × portfolio value × portfolio beta

$$\left(\frac{21{,}050}{21{,}350} - 1\right) \times €120\text{m} \times 1.3 = -€2{,}192{,}037$$

Short futures performance:

(21,315 − 21,025) = 290 points

Points × multiplier × contracts = 290 × €5 × 563 = €816,350

Return on portfolio:

−€2,192,037 + €816,350 = −€1,375,687

Return if the portfolio had a beta of 0.8:

$$\left(\frac{21{,}050}{21{,}350} - 1\right) \times €120\text{m} \times 0.8 = -€1{,}348{,}946$$

Reason for the difference:

1. Rounding in the number of futures contracts to achieve the desired beta
2. Change in basis on the futures contracts; due to convergence (when spot = futures price) at delivery, basis typically declines over the life of a futures contract

Basis = spot − futures price

Basis at initiation = 21,350 − 21,315 = 35 points (positive basis = backwardation)

Basis 12 months later = 21,050 − 21,025 = 25 points

Basis has declined on the futures, causing the hedge to be imperfect. In this case, −10 points × €5 × 563 = −€28,150.

Note that if we remove the impact of the basis change on the combined portfolio, we would get a loss of = −€1,375,687 +€28,150 = −€1,347,537. This is closer to the result if the portfolio had achieved a beta of 0.8 (the remaining difference due to rounding to the nearest complete futures contract).

(Module 17.3, LOS 17.c)

Question #20 of 74

Question ID: 1675217

A portfolio manager has a net long position in both stocks and bonds, and no cash. When attempting to replicate the existing portfolio, using bond and stock futures, which of the following statements is *most accurate*? The manager will:

A) go long on the stock futures, but short on the bond futures.

B) choose a single futures contract and net the bond and stock position.

C) go long on both stock and bond futures.

Explanation

Because the original portfolio is long in both stocks and bonds, the manager will go long on both stock and bond futures contracts.

(Module 17.3, LOS 17.c)

Question #21 of 74

Question ID: 1675200

Justin Nate wishes to speculate on the direction of interest rates, based on his belief that they will rise over the coming year. Specifically, Nate plans to use interest rate derivatives that are based on 30-day loan periods and that mature in 240 days. Current 240-day USD LIBOR is 5% and 270-day LIBOR is 5.2%. Which of the following statements is *least accurate*?

A) The futures price of the Eurodollar future is 99.45.

B) The forward price of the FRA and the forward rate implied by the Eurodollar future should be the same.

C) The current forward price for an FRA with a 30-day loan period that matures in 240 days is 6.6%.

Explanation

The forward price for a forward rate agreement (FRA) is a forward rate of interest derived from current LIBOR rates:

$$\text{Forward price rate} = \left[\frac{(1+\text{270-day LIBOR})}{(1+\text{240-day LIBOR})} - 1\right] \times \frac{360}{\text{Borrowing / Lending days}}$$

This formula, used to price an FRA, was first encountered in the Level II curriculum. The price of the FRA is the fixed rate, which is the annualized forward interest over the borrowing or lending period:

$$\text{270-day LIBOR} = 5.2 \times \frac{270}{360} = 3.9\%$$

$$\text{240-day LIBOR} = 5.0 \times \frac{240}{360} = 3.333\%$$

$$\text{Forward price (rate)} = [\frac{(1.0390)}{(1.0333)} - 1] \times \frac{360}{30} = 0.066 \text{ or } 6.6\%$$

Both the FRA and the Eurodollar future lock into a guaranteed forward rate from expiry of the contract until the end of the borrowing and lending period. The standardization of the STIR means that the borrowing and lending period lasts for 30-days after the maturity of the future. FRAs are OTC derivatives and therefore the borrowing and lending period is customized, however a FRA with a 30-day borrowing and lending period is equivalent to a STIR future. To prevent arbitrage between the two contracts the locked in forward rate should be approximately the same in the two contracts.

The STIR future is priced using IMM index convention = 100 − annualized forward rate.

Eurodollar future price = 100 − 6.60 = 93.4.

The error made in answer "The futures price of the Eurodollar future is 99.45" was a failure to annualize the forward rate.

(Module 17.1, LOS 17.a)

Question #22 - 25 of 74

Question ID: 1675228

Which of the following statements regarding the currency swap is *least accurate*?

A) Karim is taking the euro coupon payments received from his bond investments and passing those through to the swap dealer in exchange for USD. ✗

B) Currency risk is hedged in the swap. ✗

C) There is an exchange of notional amounts at the beginning and ending of the swap. ✓

Explanation

In a currency swap there may or may not be an exchange of notional amounts at the beginning and ending of the swap. Currency risk is hedged in the swap because the interest payments are based on the market reference rate (MRR) and not the exchange rate. Additionally, when the notional amounts are swapped, the amount swapped at the beginning is then returned at the end of the swap, further reducing currency risk exposure.

A cross-currency basis swap is a currency swap in which the notional principals are exchanged because the goal of the transaction is to issue at a more favorable funding rate and swap the amount to the currency of choice. In a cross-currency basis swap the non-USD currency usually trades at a negative basis to the USD, which means the party receiving the non-USD interest payment would receive the MRR less basis. The basis is the difference between interest rates in the currency swap and interest rates used to determine the forward exchange rates and refers to the number of basis points that are usually subtracted from the non-USD MRR.

(Module 17.2, LOS 17.b)

Question #23 - 25 of 74

Question ID: 1675229

Based on Karim's current and target U.S. fixed-income exposure and **Exhibit 1: Selected Information Pertaining to the Fixed-Income Futures Contracts**, to increase the portfolio's exposure and modified duration, Portobello should *most likely* purchase:

A) 502 futures contracts. ✓

B) 168 futures contracts. ✗

C) 335 futures contracts. ✗

Explanation

To solve this question, first calculate the number of futures contracts that are required to increase the portfolio's modified duration from 6 to 8. Next, calculate the number of futures contracts that are required to increase the portfolio's exposure from USD 100 million to USD 150 million.

Step 1: Increase the portfolio duration.

Noting that the modified duration of the original portfolio, which is valued at USD 100 million, needs to be increased from 6 to 8, the required number of futures contracts is calculated as:

$$\text{BPVHR} = \frac{\text{BPV}_\text{T} - \text{BPV}_\text{P}}{\text{BPV}_\text{CTD}} \times \text{CF}$$

where *BPVHR* is the number of future contracts to be bought or sold, BPV_P is the basis point value (BPV) of the portfolio, BPV_T is the target BPV of the portfolio, BPV_CTD is the BPV of the cheapest-to-deliver bond under the futures contract, and *CF* is the conversion factor.

$\text{BPV}_\text{P} = \text{MD}_\text{P} \times 0.01\% \times \text{MV}_\text{P}$, where MD_P is the current portfolio modified duration and MV_P is the market value of the portfolio. $\text{BPV}_\text{P} = 6 \times 0.01\% \times 100{,}000{,}000 = \text{USD } 60{,}000$.

$\text{BPV}_\text{T} = \text{MD}_\text{T} \times 0.01\% \times \text{MV}_\text{T}$, where MD_T is the target modified duration and MV_P is the market value of the portfolio. $\text{BPV}_\text{T} = 8 \times 0.01\% \times 100{,}000{,}000 = \text{USD } 80{,}000$.

$\text{BPV}_\text{CTD} = \text{MD}_\text{CTD} \times 0.01\% \times \text{MV}_\text{CTD}$, where MD_CTD is the modified duration of the CTD bond and

$$\text{MV}_\text{CTD} = \frac{\text{CTD price}}{100} = \frac{130}{100}.$$

$$\text{BPV}_\text{CTD} = 9 \times 0.01\% \times \frac{130}{100} \times 100{,}000 = \text{USD } 117.$$

$$\text{BPVHR} = \frac{80{,}000 - 60{,}000}{117} \times 0.98 = 168.$$

Step 2: Increase the portfolio exposure from USD 100 million to USD 150 million.

To increase the portfolio exposure, the number of future contracts to be purchased is calculated using the same formula for *BPVHR* that is illustrated previously.

$\text{BPV}_\text{P} = 0$ because the 150 million – 100 million = USD 50 million is in cash.

$\text{BPV}_\text{T} = 8 \times 0.01\% \times 50{,}000{,}0000 = \text{USD } 40{,}000$.

$\text{BPV}_\text{CTD} = 117$ (calculated previously).

$$\text{BPVHR} = \frac{40{,}000 - 0}{117} \times 0.98 = 335.$$

Thus, the total number of future contracts that need to be bought is 168 + 335 = 503.

The number of futures contracts that will increase the portfolio's exposure is omitted.

The number of futures contracts that will increase the modified duration of the original portfolio is omitted.

(Module 17.1, LOS 17.a)

Question #24 - 25 of 74

Question ID: 1675230

Based on **Exhibit 2: Selected Information Pertaining to the Equity Swap**, the expected net cash inflow on the first settlement date of the equity swap is *closest* to:

A) USD 0.59 million.

B) USD 0.14 million.

C) USD 0.44 million.

Explanation

To determine the expected cash flow on the first settlement date, calculate the expected return on the index and adjust the expected annual floating rate to a semiannual basis.

$$\text{Expected index return} = \frac{4{,}140}{4{,}200} - 1 = -1.4286\%.$$

$$\text{Expected floating rate adjusted to a semiannual basis} = \frac{1.5\%}{12} \times 6 = 0.75\%.$$

Expected net payment rate =0.75% – (–1.4286%) = 2.1786%.

Thus, the expected net cash inflow = 0.021786 × 20,000,000 = USD 0.44 million.

The USD 0.14 million answer option is incorrect as the expected net payment rate is miscalculated as 1.4286% – 0.75% = 0.6786%, resulting in the expected net cash inflow = 0.006786 × 20,000,000 = USD 0.14 million.

The USD 0.59 million answer option is incorrect as the expected floating rate is not adjusted to a semiannual basis, resulting in the expected net payment rate =1.5% – (–1.4286%) = 2.8286%, and expected net cash inflow = 20,000,000 × 0.029286 = USD 0.59 million.

(Module 17.3, LOS 17.c)

Question #25 - 25 of 74

Question ID: 1675231

Portobello's statement pertaining to a variance swap is *most likely*:

A) incorrect regarding the swap involving net payments over its life that are based on the difference between the realized variance and the variance strike.

B) incorrect regarding the swap's sensitivity to changes in implied volatility as the swap approaches its maturity.

C) correct.

Explanation

A *variance swap* is an instrument that is used for taking directional bets on implied versus realized volatility. This instrument is called a *swap* because it has a payoff that is analogous to that of a swap. However, in a variance swap, there is no exchange of cash at inception or during the life of the swap.

A variance swap is an instrument that does not involve net payments over its life.

In a variance swap, the buyer pays the difference between the fixed variance strike and the realized variance underlying the contract. If the realized variance is higher/lower than the swap's strike variance, the payoff of a long/short swap variance position is positive. Importantly, the closer the variance swap is to its maturity, the lower its sensitivity to implied volatility is.

(Module 17.4, LOS 17.d)

Question #26 of 74

Question ID: 1675207

David Lizotte wishes to completely hedge his $40m U.S. fixed-income portfolio against parallel movements in the yield curve. David is considering using the following U.S. Treasury bond future:

U.S. Treasury Bond Future	
Futures price	$154.88
Futures contract size	$100,000
CTD	3.7%, 20 years to maturity
CTD price	$113.71
CTD conversion factor	0.7342
CTD modified duration	14.51

Lizotte's fixed-income portfolio currently has a modified duration of 12. The *most* appropriate futures contract transaction to fully hedge the portfolio would be to:

A) sell 291 contracts.

B) buy 291 contracts.

C) sell 214 contracts.

Explanation

To reduce portfolio duration, Lizotte will need to sell Treasury bond futures.

Step 1: Compute the basis point value of the portfolio ($BPV_{portfolio}$):

$$BPV_{portfolio} = 12 \times 0.0001 \times \$40{,}000{,}000 = \$48{,}000$$

Step 2: Compute the basis point value of the CTD (BPV_{CTD}):

$$BPV_{CTD} = 14.51 \times 0.0001 \times [(\$113.71 / 100 \times £100{,}000] = \$164.99$$

Step 3: Compute the basis point value hedge ratio:

$$\text{BPV HR} = \frac{\text{BPV}_{\text{Target}} - \text{BPV}_{\text{Portfolio}}}{\text{BPV}_{\text{CTD}}} \times \text{CF} = \frac{\$0 - \$48{,}000}{\$164.99} \times 0.7342 = -213.6 \approx -214$$

To fully hedge the portfolio, the BPV_{target} is set equal to $0.

(Module 17.1, LOS 17.a)

Question #27 of 74

Question ID: 1675198

Jeanne Leon is the portfolio manager for a U.S. fixed income fund. Leon is concerned that central bank policy will lead to rising interest rates over the next two years. In anticipation of this, Leon wishes to reduce the duration of his portfolio. Which of the following derivatives is *most likely* to achieve his goals?

A) Series of short FRA positions. ✗

B) Two-year receiver swap. ✗

C) Short position in Treasury bond futures. ✓

Explanation

Receiver swaps (receive fixed, pay floating) can be synthetically replicated by issuing FRNs and using the proceeds to buy fixed-coupon bonds. FRNs have lower durations than fixed-coupon bonds, resulting in receiver swaps increasing duration and payer swaps reducing duration. Leon should, therefore, enter payer interest rate swaps if she wishes to reduce the portfolio's duration.

A short FRA (forward rate agreements) is an obligation to pay floating and receive fixed. A series of short FRAs replicates a receiver swap. This strategy will increase the portfolio's duration. Leon should consider a series of long FRA contracts to replicate a payer swap if the goal is to reduce portfolio duration.

Taking positions such as U.S. Treasury bond futures can be used to modify a portfolio's duration. Long futures positions increase the exposure to fixed rate returns, and short futures positions will decrease the exposure.

(Module 17.1, LOS 17.a)

Question #28 of 74

Question ID: 1675238

Michael Hallen, CFA, manages an equity portfolio with a current market value of $78 million and a beta of 0.95. Convinced the market is poised for a significant upward movement, Hallen would like to increase the beta of the portfolio by 40%, using S&P 500 futures currently trading at 856. The multiplier is 250. What is the number of futures contracts, rounded up to the nearest whole number, that will be needed to achieve Hallen's objective?

A) 144.

B) 143.

C) 139.

Explanation

First, determine the new target beta by multiplying the current beta of the portfolio, which is 0.95 by 1.4 to achieve a new target beta that is 40% greater than the current portfolio beta:

(0.95)(1.4) = 1.33

Then, use the equation:

$[(beta_T - beta_p) / beta_f][V_p / (P_f \times multiplier)]$

[(1.33 - 0.95) / 1](78,000,000) / (856)(250) = (0.38)(364.49) = 138.50, rounded to 139

(Module 17.5, LOS 17.e)

Question #29 of 74

Question ID: 1675240

Robert Zorn, CFA, manages an equity portfolio with a current market value of $150 million. The beta of the portfolio is 1.23, and Zorn is forecasting a short-term market adjustment that will significantly lower equity values and will occur in the near future. Zorn has decided to use S&P 500 futures, currently trading at 1,260, to reduce the portfolio's systematic risk exposure by 30%. The multiplier is 250. What is the number of futures contracts, rounded up to the nearest whole number, that will be needed to achieve Zorn's objective?

A) Sell 169.

B) Sell 176.

C) Buy 182.

Explanation

First, determine the new target beta by multiplying the current beta of the portfolio, which is 1.23 by 0.7, to achieve a new target beta that is 30% less than the current portfolio beta:

(1.23)(0.7) = 0.861

Then, use the equation:

$[(beta_T - beta_p) / beta_f][V_p / (P_f \times multiplier)]$

[(0.861 – 1.23) / 1](150,000,000) / (1,260)(250) = (–0.369)(476.19) = –175.71, rounded to –176.

(Module 17.5, LOS 17.e)

Question #30 of 74

Question ID: 1675203

Ashley Catrina works for a U.K.-based private equity fund. A potential new portfolio company has been identified, and the fund is considering making the acquisition. Due to cash flow timings, the fund will require a 3-month £50m bridge loan commencing in nine months' time. Catrina wishes to lock in the interest rate on the bridge loan, so she sells 3-month Sterling futures maturing in nine months at 99.25. The contract size is £500,000. In nine months' time, Catrina initiates the loan at 0.90% and closes out the hedge at 99.10. The effective interest rate on the loan is *closest* to:

A) 0.90%.

B) 0.75%.

C) 0.80%.

Explanation

No computations are required. By selling futures contracts at 99.25, Catrina is locking into a three-month interest rate of 100 − 99.25 = 0.75% in nine months' time.

There is a key point with futures contracts that have an interest-bearing instrument as the underlying asset:

Selling futures = notional borrowing

Buying futures = notional lending

The 0.75% effective interest rate on the loan can also be proven mathematically:

Catrina needs £50,000,000 / £500,000 = sell 100 contracts

In nine months' time:

Cost of loan = £50,000,000 × 0.0090 × 90 / 360 = £112,500

Profit on futures:

Profit/(loss) on contract = (forward rate purchased − forward rate sold) × 90 / 360 × contract size

Profit/(loss) on contract = (0.0090 − 0.0075) × 90 / 360 × £500,000 = £187.50

Profit on futures = 100 × £187.50 = £18,750

Net position:

Cost of loan − profit on futures = £112,500 − £18,750 = £93,750

£93,750 / £50,000,000 × 360 / 90 = 0.0075 or 0.75%

(Module 17.1, LOS 17.a)

Question #31 of 74

Question ID: 1675214

If the USD is trading at a positive basis, which of the following statements regarding cross-currency swaps is *least accurate*?

A) Additional returns can be earned on the USD relative to the interest rate suggested by covered interest rate parity. ✗

B) A non-U.S. investor can earn returns greater than their domestic return by lending their domestic currency using a cross-currency swap and investing in the U.S. debt markets. ✓

C) Covered interest parity must not hold. ✗

Explanation

If the USD is trading at a positive basis, it means that the USD interest rate in a cross-currency swap is greater than the $ interest rate used to price the forward currency rate using covered interest rate parity (CIRP). This means that lenders of the dollar can generate additional returns lending dollars using a cross-currency swap.

CIRP suggest that, due to the activities of arbitragers, no such premium should exist: basis, if CIRP holds, should be zero. The existence of positive basis on the dollar against most currencies, therefore, indicates CIRP is not holding. The additional costs of arbitrage due to regulatory changes are suggested as the reason for the failure of CIRP. Changes to capital adequacy requirements in particular have led to lower levels of arbitrage in the markets. The reason the USD has positive basis (and, therefore, most nondollar currencies show –ve basis) is the high demand for $ loanable funds.

If the USD shows positive basis, U.S. investors can use cross-currency swaps to lend USD and borrow overseas currency and then invest in overseas debt markets. The currency swap allows the U.S. investor to convert their returns back into $ at fixed exchange rates and earn the positive basis. This, theoretically, will generate higher returns (via the basis) than investing domestically.

(Module 17.2, LOS 17.b)

Question #32 of 74

Question ID: 1675202

Eagle Jones purchased Eurodollar interest rate futures nine months ago. The contracts have now matured.

Contract size	$1,000,000
Nine-month futures price (at initiation)	96.45
Number of contracts purchased	10
Three-month USD MRR at futures' maturity	4.5%

Using the contract details, the profit on Jones's position at contract maturity is *closest* to:

A) −$9,500. ✗

B) +$2,400. ✗

C) −$23,750. ✓

Explanation

A futures price of 96.45 is equivalent to a forward interest rate of 3.55% at contract expiry.

Profit on contract = (forward rate purchased − forward rate sold) × 90 / 360 × contract size

(0.0355 − 0.045) × 90 / 360 × $1,000,000 = −$2,375 per contract

10 × −$2,375 = −$23,750

(Module 17.1, LOS 17.a)

Question #33 of 74

Question ID: 1675197

A company has issued a 5-year semiannual-pay FRN with coupons equal to £LIBOR + 60 bp, with a par value of £50m. The company enters a 5-year semiannual interest rate swap, with notional principal of £50m, as the fixed-rate payer. The fixed rate in the swap is 4%. The 180-day £LIBOR at Day 360 is 5%. What is the net interest cost to the firm at the third settlement date on the swap?

A) £1,400,000. ✗

B) £1,150,000. ✓

C) £2,300,000. ✗

Explanation

Interest on FRN:

£50m × (5% + 0.60%) × 180 / 360 = £1,400,000

Payer swap:

Pay fixed:

£50m × 4% × 180 / 360 = £1,000,000

Receive floating:

£50m × 5% × 180 / 360 = £1,250,000

Net receipt on swap:

£1,250,000 − £1,000,000 = £250,000

Net interest cost:

£1,400,000 − £250,000 = £1,150,000

Note that the company has locked in a fixed cost equal to the fixed rate on the swap plus the basis points above LIBOR on the FRN:

£50,000,000 × (4% + 0.60%) × 180 / 360 = £1,150,000.

(Module 17.1, LOS 17.a)

Question #34 of 74

Question ID: 1675199

Andrew Jeremy is a U.K. fixed income manager who wishes to use interest rate swaps to increase the duration of his bond portfolio. The existing portfolio has a market value of £20m and a modified duration of 7. Jeremy wishes to increase the modified duration of his portfolio to 9 for the next two years. Two-year payer swaps currently have a duration of −5. What type of swap and what amount of notional principal will Jeremy require to achieve his stated goal?

A) Receiver swap notional principal £20m.

B) Receiver swap notional principal £8m.

C) Payer swap notional principal £20m.

Explanation

Payer swaps (pay fixed, receive floating) are the synthetic equivalent of issuing (shorting) fixed-income securities and purchasing (long) FRNs. FRNs have a duration approximately equal to the time in years to the next coupon reset date. Payer swaps have overall negative durations.

Receiver swaps being the equivalent of issuing FRNs and purchasing fixed-coupon bonds will increase a portfolio's duration. Note that the duration is equal and opposite that of the payer swap. The payer swap = −5; therefore, the receiver swap = +5.

$$NP_S = \left(\frac{MD_T - MD_P}{MD_S}\right)(MV_P)$$

where:

NP_S = notional swap principal

MD_T = target modified duration

MD_P = current portfolio modified duration

MD_S = modified duration of swap

MV_P = market value of portfolio

$$NP_S = \left(\frac{9-7}{5}\right)(£20m) = £8m$$

(Module 17.1, LOS 17.a)

Question #35 of 74

Question ID: 1675239

An investor has an $80 million diversified stock portfolio with a beta of 1.1. He would like to partially hedge his portfolio using S&P 500 futures contracts. The contracts are currently trading at $596.70 with a multiple of $250. Which of the following trades will reduce the portfolio beta by 50%?

A) Buy 295 contracts.

B) Sell 590 contracts.

C) Sell 295 contracts.

Explanation

The number of futures contracts required for the 100% risk-minimizing hedge (or to reduce the beta to zero) is computed as follows:

> Number of contracts = portfolio value / futures contract value × (target beta – current beta) = $80 million / ($596.70 × $250) × (0 – 1.1) = –590 contracts (sell contracts because the amount is negative)

Therefore, to reduce the portfolio beta by 50%, we simply use half this number of contracts, or 295 contracts.

(Module 17.5, LOS 17.e)

Question #36 of 74

Question ID: 1675250

Redden Capital Management manages an intermediate, high-quality bond portfolio with a value of $12 million and a modified duration of the portfolio is 4.4 years. Scott Stuart, the manager of the portfolio, is concerned about rising interest rates over the next few months and wants to make a tactical adjustment and cut the duration of the portfolio in half. Stuart asks Amy Swemba, a junior portfolio manager with Redden, to accomplish this task. Swemba is aware that a Treasury bond futures contract exists with a value of $102,000, with a modified duration of 8.2 years. Swemba replies to Stuart's comments with the following statements:

Statement 1:	The fastest and most cost-effective way to reduce the duration of the portfolio by half would be to sell $6 million dollars' worth of the actual bonds in the portfolio.
Statement 2:	The portfolio's duration could also be adjusted by selling 40 of the Treasury bond futures contracts.

After listening to Swemba's statements, Stuart should:

A) agree with Statement 1, but disagree with Statement 2. ✗

B) disagree with Statement 1, but agree with Statement 2. ✗

C) disagree with both Statement 1 and Statement 2. ✓

Explanation

Note: On the exam, it is very likely for material on tactical asset allocation to be tested in conjunction with material from derivatives as tactical asset allocation can be accomplished by selling assets, or with a derivative overlay. Stuart should disagree with both of Swemba's statements. Although Stuart's goal of reducing the duration could be accomplished by selling bonds in the portfolio, doing so would likely incur significant transaction costs. Also, because the duration of each bond in the portfolio is likely different, specific bonds would have to be selected to accomplish Stuart's goal, making the process more difficult. A derivative overlay, accomplished by using futures contracts, would be much easier and cost effective. Swemba is also incorrect with respect to the number of futures contracts that would need to be sold. The correct number of futures contracts to be sold is:

$$BPVHR = [(BPV_{target} - BPV_{portfolio}) / BPV_{CTD}] \times CF$$

Because the futures price is given, this is used instead of $(1 / BPV_{CTD}) \times CF$

$$BPV_{target} = \$12M \times 0.0001 \times 2.2 = 2,640$$

$$BPV_{portfolio} = \$12M \times 0.0001 \times 4.4 = 5,280$$

$$BPV_{futures} = 102,000 \times 0.0001 \times 8.2 = 83.64$$

BPVHR = (2,640 – 5,280) / 83.64 = –31.56 ≈ –32 futures contracts. The minus sign means that 32 contracts should be sold to achieve the desired duration in the portfolio.

(Module 17.5, LOS 17.f)

Question #37 of 74

Question ID: 1675254

An investor has a $5 million investment in small-cap stocks. The investor enters into an equity index swap where the investor pays the return on the Russell 2000 and receives the return on the Dow Jones Industrial Average. The notional principal of the swap is $1 million. The resulting position is a synthetic mix of:

A) 20% large stocks and 80% small stocks. ✓

B) 25% large stocks and 75% small stocks. ✗

C) 16.67% large stocks and 83.33% small stocks. ✗

Explanation

After the swap, $1 million, or 20% of the portfolio's exposure will be invested in the Dow Jones Industrial Average index of large stocks—with $4 million, or 80% of the portfolio, remaining invested in small stocks. The $1 million notional principal represents 20% of the position. That is the amount that has been synthetically transferred from one class of assets to the other.

(Module 17.5, LOS 17.f)

Question #38 of 74

Question ID: 1675208

Samantha Holly is the fund manager of a German fixed-income fund. The fund has a market value of €90m and a modified duration of 14. Holly believes that the European central bank will end its policy of lower target interest rates and would like to reduce the duration of her portfolio. Holly has identified the following Bund future, which she would like to use:

Government Bond Futures	German Bund
Contract size	€100,000
Future price	€174.50
CTD price	€104.83
CTD conversion factor	0.5955
CTD modified duration	15.81

How many contracts will Holly need to sell to achieve a target duration of 8?

A) Sell 194 contracts.

B) Sell 326 contracts.

C) Sell 453 contracts.

Explanation

Holly wishes to achieve a target duration of 8 for her portfolio.

Step 1: Compute the basis point value of the portfolio ($BPV_{portfolio}$):

$$BPV_{portfolio} = 14 \times 0.0001 \times €90{,}000{,}000 = €126{,}000$$

Step 2: Compute the basis point value of the target duration:

$$BPV_{Target} = 8 \times 0.0001 \times €90{,}000{,}000 = €72{,}000$$

Step 3: Compute the basis point value of the CTD (BPV_{CTD}):

$$BPV_{CTD} = 15.81 \times 0.0001 \times [(€104.83/100 \times €100{,}000] = €165.74$$

Step 4: Compute the basis point value hedge ratio:

$$\text{BPV HR} = \frac{BPV_{Target} - BPV_{Portfolio}}{BPV_{CTD}} \times CF = \frac{€72{,}000 - €126{,}000}{€165.74} \times 0.5955 = -194.02 \approx -194$$

Holly will need to sell 194 Bund futures to achieve her target duration.

(Module 17.1, LOS 17.a)

Question #39 of 74

Question ID: 1675209

Sarah Finley wishes to hedge her portfolio of U.S. bonds against a rise in U.S. interest rates. Her portfolio has a weighted average maturity of 15 years. Sarah would *most appropriately* choose to:

A) enter into a receiver interest rate swap.

B) sell a strip of Eurodollar futures.

C) sell bond futures.

Explanation

The choice of hedging a portfolio using Eurodollar futures or fixed-income futures depends on the maturity of the portfolio. Typically, Eurodollar futures are used to hedge positions with no more than two to three years remaining to maturity. The reason for using Eurodollar futures is that they have greater liquidity than bond futures. As the maturity (tenor) of the Eurodollar futures extends, liquidity levels tend to fall. Hedging with interest rate futures requires strips of Eurodollar futures to cover multiple coupon payments. The sensitivity of each coupon to changing rates will need to be assessed when calculating the number of contracts needed in the strip.

Finley's portfolio has a weighted average maturity of 15 years; therefore, it is more likely that she will use fixed-income futures to hedge the exposure. Bond futures will typically have better liquidity for longer tenors than Eurodollar futures and allow the hedge to be constructed with greater ease than Eurodollar futures.

A receiver swap (pay floating and receive fixed) will have a positive duration. The receiver swap is synthetically the equivalent of issuing floating-rate notes and investing in fixed-coupon bonds. A receiver swap will, therefore, increase the portfolio's sensitivity to interest rates. Payer swaps have negative durations and can be used to reduce interest rate exposure.

(Module 17.1, LOS 17.a)

Question #40 of 74

Question ID: 1675223

Which of the following statements regarding the VIX Index is *most accurate*?

A) VIX is a measure of implied volatility derived from lookback options traded on the S&P 500.

B) VIX measures realized 3-month volatility on the S&P 500 Index.

C) VIX is the annualized standard deviation of the expected percentage changes in the index with 68% confidence.

Explanation

The CBOE volatility index (VIX) is a measure of the weighted average volatility computed from both 30-day call and put options on the S&P 500 Index.

If VIX has a value of 40, we can interpret this as the market, implying that the S&P 500 will stay within a range of +/− 40% over one year with a 68% level of confidence. This implies a range +/− $\frac{40}{\sqrt{12}}$ = 11.55% over the next 30-day period.

Importantly, empirical studies have shown a negative correlation between the VIX and stock returns, which becomes more pronounced during downturns. This correlation allows derivatives based on the index to offset the losses on an equity portfolio when volatility increases.

(Module 17.4, LOS 17.d)

Question #41 of 74

Question ID: 1675216

Which of the following statements about portfolio hedging is *least accurate*?

A) To synthetically create the risk/return profile of an underlying common equity security, buy the corresponding futures contract, sell the common short, and invest in a T-bill. ✓

B) For a fixed portfolio insurance horizon, using put options generally requires less rebalancing and monitoring than with the use of futures contracts. ✗

C) Futures contracts have a symmetrical payoff profile. ✗

Explanation

To synthetically create the risk/return profile of an underlying common equity security, buy the corresponding futures contract and invest in a T-bill.

Futures contracts involve margin, so there may be margin calls that require rebalancing. Option contracts do not have margin calls.

Futures contracts are a zero-sum game, to the extent that one side loses, the other side gains by the same. Therefore, the payoff profile is symmetrical.

(Module 17.3, LOS 17.c)

Question #42 of 74

Question ID: 1675253

An investor has a cash position currently invested in T-bills but would like to "equitize" it by using S&P futures contracts. Which of the following trades will create the desired synthetic equity position?

A) Selling S&P 500 futures contracts short. ✗

B) Buying S&P 500 futures contracts. ✓

C) Selling the T-bills and buying S&P 500 futures contracts. ✗

Explanation

The trader can buy stock index futures and hold them in conjunction with T-bills to mimic a stock portfolio. So, we have the following:

Synthetic stock portfolio = T-bills + stock index futures.

(Module 17.5, LOS 17.f)

Question #43 of 74

Question ID: 1675226

David Climo purchased a 12-month variance swap on the S&P 500 with a vega notional of $60,000 and a strike of 6%. Seven months have passed, and the S&P has a realized volatility of 6.5% over this period. The strike of a variance swap with five months to maturity is quoted at 5.8% at this time, and the 5-month USD LIBOR is 3%. The current value of the swap is *closest* to:

A) $31,250. ✗

B) $13,300.

C) $13,136.

Explanation

Step 1: Compute the expected variance at maturity =

$$(42.25 \times \frac{7}{12}) + (33.64 \times \frac{5}{12}) = 24.65 + 14.02 = 38.66.$$

where: $6.5^2 = 42.25$ and $5.8^2 = 33.64$

Note that expected variance is simply the time-weighted average of realized variance to date and implied variance over the remaining life of the contract.

Step 2: Compute the expected payoff at maturity.

$$\text{variance notional} = \frac{\text{vega notional}}{2 \times K} = \frac{\$60{,}000}{2 \times 6} = \$5{,}000$$

$$\text{expected payoff at maturity} = (\sigma^2 - K^2) \times \text{variance notional}$$

where: $K^2 = 6^2 = 36$

expected payoff at maturity = (38.66 − 36) × $5,000 = $13,300

Step 3: Discount expected payoff from maturity to valuation date (five months).

$$\text{unannualize the interest rate} = 3\% \times \frac{5}{12} = 1.25\%$$

$$\text{current value of swap} = \frac{\$13{,}300}{1.0125} = \$13{,}136$$

This is a gain to David (the purchaser).

(Module 17.4, LOS 17.d)

Question #44 - 47 of 74

Question ID: 1675259

Based on **Exhibit 1: Selected Information Pertaining to the Floating-Rate Bond Portfolio and the Interest Rate Swap**, the swap that Smith implemented to hedge against a potential decrease in interest rates is *most likely*:

A) a receiver swap with a notional principal of USD 92 million.

B) a payer swap with a notional principal of USD 100 million.

C) a receiver swap with a notional principal of USD 313 million.

Explanation

To hedge a portfolio of floating-rate bonds against a decrease in interest rates, the endowment can enter a receiver interest rate swap, wherein it will receive the fixed rate cash flows and pay the floating-rate cash flows. Entering the receiver swap will increase the portfolio's modified duration.

The notional principal of the swap is calculated as:

$$N_S = \frac{MD_T - MD_P}{MD_S} MV_P$$

where N_S is the notional principal of the interest rate swap, MD_T is the target modified duration of the portfolio, MD_p is the current modified duration of the portfolio, MD_S is the modified duration of the swap, and MV_p is the market value of the bond portfolio:

$$\frac{5 - 0.4}{2.5} \times 50 \text{ million} = \text{USD } 92 \text{ million}$$

Using a payer swap will not hedge the portfolio against a decrease in interest rates and will result in a decrease in the portfolio's modified duration. The notional principal is also miscalculated as

$$\frac{5}{2.5} \times 50 \text{ million} = \text{USD } 100 \text{ million.}$$

The notional principal is miscalculated as $\frac{5 - 2.5}{0.4} \times 50 \text{ million} = \text{USD } 312.5 \text{ million.}$

(Module 17.1, LOS 17.a)

Question #45 - 47 of 74

Question ID: 1675260

Upon receiving the $500 million in grants, if the 90-day LIBOR is 2%, the effective interest received on the grant portfolio is *closest* to:

A) USD 3,000,000. ✗

B) USD 2,625,000. ✗

C) USD 3,125,000. ✓

Explanation

The effective interest received on the grant portfolio is calculated as:

effective interest = interest received on the deposit + profit on the futures contracts

Noting that the 90-day LIBOR in 2 months is 2%, the interest received on the deposit is calculated as:

$$500 \text{ million} \times (2\% + 0.1\%) \times \frac{90}{360} = \text{USD } 2,625,000$$

By buying the Eurodollar futures contracts at 97.6, Smith locked an interest rate of 1 – 0.976 = 2.4%. Because the 90-day LIBOR prevailing in two months (2%) is lower than the locked interest rate, Smith generated a profit on the futures contracts. This profit is calculated as:

profit on the Eurodollar futures contracts = –difference in interest rates (in bps) × number of contracts bought × BPV of each contract

profit on the futures contract = –(200 – 240) × 500 × 25 = USD 500,000

So, effective interest = 2,625,000 + 500,000 = USD 3,125,000.

The USD 2,625,000 answer option is incorrect as the profit on the Eurodollar futures contracts is not considered.

The USD 3,000,000 answer option is incorrect as the extra 10 bps received on the USD 500 million deposit is not considered.

(Module 17.1, LOS 17.a)

Question #46 - 47 of 74

Question ID: 1675261

Which suggestion will *most likely* hedge the EUR-denominated investment funded by the $600 million in grants received from the international NGO?

A) Suggestion 2. ✗

B) Suggestion 1. ✓

C) Suggestion 3. ✗

Explanation

Suggestion 1 is correct. To invest in the European bonds, the USD 600 million received in grants should be converted into EUR. The endowment will later convert the semiannual interest payments received on the bonds to USD. Accordingly, a pay EUR, receive USD currency swap with semiannual settlements is suitable for hedging against a depreciation of EUR against USD. The currency swap will exchange interest payments on a semiannual basis and should involve exchanging notional principals at both initiation and maturity.

Suggestion 2 will increase the exposure to EUR instead of hedging it.

Suggestion 3 will only hedge the principal of the bonds—it will not hedge the interim EUR coupons received.

(Module 17.2, LOS 17.b)

Question #47 - 47 of 74

Question ID: 1675262

Based on **Exhibit 2: Selected Information Pertaining to the S&P Equity Futures Contract**, to equitize the excess cash balance and increase the portfolio's exposure to the S&P 500, Smith should *most likely*:

A) short 17 futures contracts on the S&P 500 Index.

B) long 17 futures contracts on the S&P 500 Index.

C) long 4,250 futures contracts on the S&P 500 Index.

Explanation

The number of futures contracts required to equitize the excess cash portfolio is calculated as:

$$N_f = \left(\frac{\beta_T}{\beta_f}\right)\left(\frac{S}{F}\right)$$

where N_f is the number of futures contracts, β_T is the portfolio's target beta, β_f is the futures contract's beta, S is the size of the portfolio, and F is the actual futures price, which is equal to the futures contract multiplier × futures contract price:

$$F = 250 \times 3{,}500 = \$875{,}000$$

Because Smith would like to establish an exposure to the S&P 500, $\beta_T = 1$

$$N_f = \frac{1}{1} \times \frac{14{,}875{,}000}{875{,}000} = 17$$

Thus, Smith will have to purchase 17 future contracts.

A short position (i.e., the short 17 futures contracts answer option) will not increase the portfolio's beta and achieve the required exposure to the S&P 500 Index.

With the 4,250 futures contracts answer option, the number of futures contracts is miscalculated as:

$$\frac{1}{1} \times \frac{14{,}875{,}000}{3{,}500} = 4{,}250$$

(Module 17.5, LOS 17.e)

Question #48 of 74

Question ID: 1675220

Sofia Chiara manages an equity fund that invests in Italian stocks. The fund has a market value of €120,000,000 and a beta of 1.3 relative to the major Italian index (FTSE MIB). Chiara wishes to reduce the beta of her portfolio to 0.8 over the next 12 months. Chiara will use the FTSE MIB future with 24 months to maturity to achieve her goal. Contract details are as follows:

FTSE MIB Future	
24 months futures contract price	21,312
Multiplier	€5
Current index value	21,350

The futures position that Chiara will need to take to achieve her desired beta is *closest* to:

A) short 11 contracts.

B) short 2,815 contracts.

C) short 563 contracts.

Explanation

Be careful when answering questions containing index futures. Futures contracts will often have different multipliers. The S&P 500 Index has a multiplier of $250. If you used that multiplier, you would arrive at one of the other answer choices. The FTSE MIB has a multiplier of €5. In any case, the correct multiplier will always be given in the case facts.

Short futures positions are used to decrease the beta of a portfolio, and long futures positions are used to increase the beta of a portfolio. A portfolio's beta is the weighted average of the stocks' betas contained within the portfolio.

$$\text{Number of futures required} = \left(\frac{\beta_T - \beta_P}{\beta_F}\right)\left(\frac{MV_P}{F}\right)$$

where:

β_T = target portfolio beta

β_P = current portfolio beta

β_F = futures beta (beta of stock index)

MV_P = market value of portfolio

F = futures contract value = futures price × multiplier

Number of futures required = (0.8 − 1.31)[€120,000,000 / (21.312 × €5)]= −563.06

Chiara will need to sell 563 futures contracts to achieve her target portfolio beta.

(Module 17.3, LOS 17.c)

Question #49 of 74

Question ID: 1675248

An investor has a $100 million stock portfolio with a beta of 1.2. He would like to alter his portfolio beta using S&P 500 futures contracts. The contracts are currently trading at 596.90. The futures contract has a multiple of 250. Which of the following is the correct trade required to double the portfolio beta?

A) Sell 804 contracts.

B) Buy 1,608 contracts.

C) Buy 804 contracts.

Explanation

The number of futures contracts required to double the portfolio beta is computed as follows:

> Number of contracts = [(target beta – portfolio beta) / futures beta] × (portfolio value / futures contract value) = [(2.4 – 1.2) / 1] × [$100 million / (596.90 × $250)] = 804 contracts.

To double the portfolio beta, we buy 804 contracts.

(Module 17.5, LOS 17.f)

Question #50 - 53 of 74

Question ID: 1675269

Based on Exhibit 1 and the information pertaining to the growth and the value portfolios, to temporarily shift 40% of the value portfolio's assets to the growth portfolio, Arniston should *most likely*:

A) short 26,087 futures contracts on value stocks and long 18,095 futures contracts on growth stocks. ✗

B) short 87 futures contracts on value stocks and long 60 futures contracts on growth stocks. ✓

C) short 18 futures contracts on value stocks and long 16 futures contracts on growth stocks. ✗

Explanation

To temporarily shift 40% of the AUD 25 million from the value portfolio to the growth portfolio, Arniston should buy futures contracts on growth stocks and sell futures contracts on value stocks:

value of assets to shift = 40% × 25 million = AUD 10 million

The number of futures contracts is calculated as:

$$\text{number of contracts} = \left(\frac{\beta_T - \beta_S}{\beta_f}\right)\left(\frac{S}{F}\right)$$

where β_T is the target beta of the portfolio, β_s is the beta of the current portfolio, β_f is the beta of the futures contract, *S* is the size of the portfolio, and *F* is the actual futures price, which is calculated as quoted futures price × futures contract multiplier:

$$\text{number of contracts}_{\text{value}} = \left(\frac{0-1.20}{1.15}\right)\left(\frac{10{,}000{,}000}{300\times400}\right) = -87$$

$$\text{number of contracts}_{\text{growth}} = \left(\frac{0.95-0}{1.05}\right)\left(\frac{10{,}000{,}000}{300\times500}\right) = -60$$

Number of contracts$_{\text{value}}$ is miscalculated as $\left(\frac{0.95-1.20}{1.15}\right)\left(\frac{10{,}000{,}000}{300\times400}\right) = -18.$

Number of contracts$_{\text{growth}}$ is miscalculated as $\left(\frac{1.2-0.95}{1.05}\right)\left(\frac{10{,}000{,}000}{300\times500}\right) = 16.$

The multiplier for both futures contracts is omitted, resulting in the following:

Number of contracts$_{\text{value}}$ = $\left(\frac{0-1.2}{1.15}\right)\left(\frac{10{,}000{,}000}{400}\right) = -26{,}087$

Number of contracts$_{\text{growth}}$ = $\left(\frac{0.95-0}{1.05}\right)\left(\frac{10{,}000{,}000}{500}\right) = 18{,}095$

(Module 17.5, LOS 17.e)

Question #51 - 53 of 74

Question ID: 1675270

The effective beta of the U.S. stock portfolio after using the five futures contracts is *closest* to:

A) 1.01. ✗

B) 0.66. ✗

C) 1.24. ✓

Explanation

The profit on the U.S. stock portfolio includes the profit on the futures contracts and the profit on the original portfolio.

To increase the beta of the initial position, Arniston should purchase futures contracts. Because the index increased in value, the long position in the futures contracts generated a profit that is equal to 250 × 5 × (3,570 – 3,500) = USD 87,500.

Because the original portfolio has a beta of 0.95 and the market return is 2%, the return on the portfolio is 0.95 × 2% = 1.9%. Thus, the profit on the original portfolio at the end of the period is 15,000,000 × 1.019 = USD 15,285,000.

Total portfolio value at end of period = 15,285,000 + 87,500 = USD 15,372,500.

$$\text{Portfolio return} = \frac{15{,}372{,}000}{15{,}000{,}000} - 1 = 0.0248 = 2.48\%.$$

$$\text{Effective beta of the portfolio} = \frac{\text{portfolio return}}{\text{broad market return}} = \frac{2.48\%}{2\%} = 1.24.$$

The profit on the futures contracts is miscalculated as 250 × 5 × (3,500 – 3,570) = –USD 87,500, resulting in:

Total portfolio value = 15,285,000 – 87,500 = USD 15,197,500

$$\text{Portfolio return} = \frac{15{,}197{,}500}{15{,}000{,}000} - 1 = 1.32\%$$

$$\text{Effective beta of the portfolio} = \frac{1.32}{2} = 0.66$$

The profit on the futures contracts is miscalculated because the number of futures contracts is omitted, resulting in 250 × (3,570 – 3,500) = USD 17,500. Then, this results in:

Total portfolio value = 15,285,000 + 17,500 = USD 15,302,500

$$\text{Portfolio return} = \frac{15{,}302{,}500}{15{,}000{,}000} - 1 = 2.02\%$$

$$\text{Effective beta of the portfolio} = \frac{2.02\%}{2\%} = 1.01$$

(Module 17.3, LOS 17.c)

Question #52 - 53 of 74

Question ID: 1675271

Based on Exhibit 2, the probability of a 30 bp hike in the fed funds rate is *closest* to:

A) 0.66.

B) 0.33.

C) 0.58.

Explanation

The probability of a 30 bp hike in the fed funds rate is calculated as:

$$\frac{\text{effective federal funds rate} - \text{current federal funds rate}}{\text{federal funds rate assuming a rate hike} - \text{current federal funds rate}}$$

The effective fed funds rate is implied by the fed funds futures price, which is calculated as 100% – 98.8% = 1.2%.

The current fed funds rate is the average of the current fed funds rate target range, which is calculated as:

$$\frac{0.95\%+1.10\%}{2} = 1.025\%$$

By adding 30 bps to the lower and upper bounds of the current fed funds target range, the fed funds rate target range (assuming a hike of 30 bps) is obtained. As such:

$$\text{federal funds rate assuming a 30 bp rate hike} = \frac{(0.95\%+0.30\%)+(1.10\%+0.30\%)}{2} = 1.325\%$$

probability of a Fed funds rate hike = $\frac{1.20\%-1.025\%}{1.325\%-1.025\%} = 0.58$

The 0.33 answer option is incorrect as the probability of a fed funds rate hike is miscalculated as $\frac{1.2\%-1.1\%}{1.4\%-1.1\%} = 0.33.$

The 0.66 answer option is incorrect as the probability of a fed funds rate hike is miscalculated as $\frac{1.4\%-1.2\%}{1.4\%-1.1\%} = 0.66.$

(Module 17.5, LOS 17.f)

Question #53 - 53 of 74

Question ID: 1675272

Noting the market conditions mentioned in Arniston's note and assuming she buys futures contracts on VIX, her return on the futures contracts will *most likely*:

A) decrease given the negative roll yield.

B) increase given the positive roll yield.

C) not be affected by the roll yield.

Explanation

Because investors are expecting volatility to be higher in the short term than in the long term, VIX futures prices are in backwardation, and the VIX futures price will be pulled to the spot rate the closer the contract approaches expiry (the contracts will increase in price the closer they get to maturity). Accordingly, Arniston, who is long a VIX futures contract, will realize a positive roll yield that will increase the return on the futures contract.

Because futures prices are in backwardation (not in contango), the roll yield will be positive and thereby increase the total return on the futures contract.

(Module 17.4, LOS 17.d)

Question #54 of 74

Question ID: 1675257

Alan Carney manages a fixed-income fund and is concerned that the Federal Open Market Committee will increase the target rate. Alan has collected the following data:

Current fed funds target	1.75%−2.00%
Fed funds futures price*	98

*Nearest futures contract maturity after the date of the next FOMC meeting

The implied probability of a 25 bp increase in the target rate at the next FOMC meeting is *closest* to:

A) 50%.

B) 12.5%.

C) 25%.

Explanation

To determine the probability of a change in the fed funds target rate:

$$\%\text{ probability of rate change} = \frac{\text{effective rate implied by futures} - \text{current fed funds target rate}}{\text{fed funds rate assuming a rate change} - \text{current fed target funds rate}}$$

Expected average effective rate at contract expiration = 100 − 98.00 = 2.0%

$$\text{Current target rate midpoint} = \frac{1.75\% + 2.00\%}{2} = 1.875\%$$

Target rate assuming a 25 bp rise = 1.875% + 0.25% = 2.125%

$$\%\text{ probability of a rate change} = \frac{2.00\% - 1.875\%}{2.125\% - 1.875\%} = \frac{0.125\%}{0.25\%} = 50\%$$

(Module 17.5, LOS 17.f)

Question #55 of 74

Question ID: 1675244

A manager of a $20,000,000 portfolio wants to decrease beta from the current value of 0.9 to 0.5. The beta on the futures contract is 1.1 and the futures price is $105,000. Using futures contracts, what strategy would be appropriate?

A) Short 69 contracts.

B) Short 19 contracts.

C) Long 69 contracts.

Explanation

Number of contracts = -69.26 = (0.5 − 0.9) × ($20,000,000) / (1.1 × $105,000), and this rounds down to 69 (absolute value). Since the goal is to decrease beta, the manager should go short which is also indicated by the negative sign.

(Module 17.5, LOS 17.e)

Question #56 of 74

Question ID: 1675195

A U.S. firm that borrows dollars and uses a plain vanilla currency swap to obtain euros for an investment in Europe is *most likely* trying to:

A) lower borrowing costs. ✓

B) increase the duration of the position. ✗

C) create a synthetic pay-fixed dollar loan. ✗

Explanation

Swaps can lower overall borrowing costs by allowing firms to borrow at a lower rate within their own country rather than paying a higher rate by borrowing directly in the foreign currency. For example, a U.S. borrower needing euros would have to pay a higher rate than a counterparty in Europe. The European counterparty can borrow at a lower rate and pass the savings on to the U.S. borrower, who passes similar savings back via borrowing dollars in the United States and exchanging them for the euros. None of the other answers make sense.

(Module 17.1, LOS 17.a)

Question #57 of 74

Question ID: 1675215

If the value of a stock portfolio equals 16 times the futures price of the appropriate equity index contract and beta of the equity portfolio, and futures prices were equal, how many contracts would it take to reduce the beta of the equity index to zero?

A) A long position in 4 contracts. ✗

B) A long position in 16 contracts. ✗

C) A short position in 16 contracts. ✓

Explanation

Number of contracts = –16 = (0 − beta) × (16 × futures price) / (beta × futures price)

(Module 17.3, LOS 17.c)

Question #58 of 74

Question ID: 1675219

Marjorie Scarlett manages a U.K. equity portfolio with a market value of £20m and a beta relative to the FTSE 100 of 1.2. Scarlett wishes to increase her exposure to U.K. equities, so she takes the equity return receiver position in a 2-year equity index swap with a semiannual settlement. The swap is based on the FTSE 100, and it has a notional principal of £5m. The FTSE 100 at the start of the swap's life is 7,520, and 6-month £ LIBOR is 4%. At the time of the first settlement date, the FTSE 100 is at 7,600 and 6-month LIBOR is 4.2%. Assuming that Scarlett's existing portfolio's beta remains unchanged at 1.2, the £ return on the portfolio at the first settlement date will be *closest* to:

A) £230,510.

B) £208,510.

C) £255,319.

Explanation

Return on original portfolio:

Opening value of portfolio × FTSE 100 return × portfolio beta

$$£20\text{m} \times \left(\frac{7{,}600}{7{,}520} - 1\right) \times 1.2 = £255{,}319$$

Net return on equity swap:

Pay floating interest:

£5m × 0.04 × 180 / 360 = £100,000

Note that LIBOR at the start of the period determines the payment at the end of the period.

Receive equity return:

$$£5\text{m} \times \left(\frac{7{,}600}{7{,}520} - 1\right) = £53{,}191$$

Net return on swap = £53,191 − £100,000 = −£46,809

Total £ return at settlement date:

Portfolio return + net return on swap = £255,319 − £46,809 = £208,510

(Module 17.3, LOS 17.c)

Question #59 of 74

Question ID: 1675196

A company has substantial bank loans and pays interest at LIBOR +80 bp. The central bank has signaled its intention to increase target interest rates over the next 12-month period. What type of interest derivatives would the company *most appropriately* enter into to mitigate its floating rate exposure?

A) A payer swap.

B) Long short-term interest rate futures.

C) Short FRA contracts.

Explanation

Short FRA positions will lose value when interest rates rise. As interest rates rise, the fixed rate that the short party receives looks less attractive as the fixed rate in new FRAs, covering the same notional borrowing and lending period, increases.

A payer swap is an agreement to pay fixed and receive floating. As interest rates rise, the floating payments received increase in size relative to the constant fixed payments. With a properly constructed interest rate swap, the net receipt on the payer swap will offset the increased interest payments on the company's bank debt.

Long short-term interest rate futures will decrease in value as interest rates rise. The pricing convention (IMM index convention) used for the future is 100 − 30-day annualized forward interest rate at maturity. As interest rates rise and the forward rate increases, the futures price will decline.

(Module 17.1, LOS 17.a)

Question #60 of 74

Question ID: 1675241

An S&P 500 Index manager knows that he will have $60,000,000 in funds available in three months. He is very bullish on the stock market and would like to hedge the cash inflow using S&P 500 futures contracts. The S&P 500 futures contract stands at 1,100.00, and one contract is worth 250 times the index. Which of the following is the *most accurate* hedge for this portfolio?

A) Sell 218 contracts. ✗

B) Buy 284 contracts. ✗

C) Buy 218 contracts. ✓

Explanation

To be hedged against stock price increases, S&P 500 futures contracts have to be purchased. The quantity of contracts to buy is computed as follows:

contracts = (beta)(portfolio value) / (futures price)(contract multiplier)

= (1)(60,000,000) / (1,100)(250) ≅ 218.18 = 218 contracts

(Module 17.5, LOS 17.e)

Question #61 - 64 of 74

Question ID: 1675233

Based on **Exhibit 1: Selected Information Pertaining to the Bonds That Can Be Delivered Under the Futures Contract**, which of the three bonds should *most likely* be purchased to be delivered by Xenjou at the maturity of the futures contract?

A) Bond C. ✗

B) Bond A. ✓

C) Bond B. ✗

Explanation

Xenjou should choose the cheapest-to-deliver bond—namely, the bond that maximizes the difference between the amount received and the amount paid at maturity of the futures contract.

First, calculate each bond's principal invoice amount as:

$$\text{principal invoice amount} = \frac{\text{futures settlement price}}{100} \times \text{CF} \times \text{size of the futures contract}$$

The principal invoice amount of Bond A $= \frac{138}{100} \times 0.9910 \times 500,000 = \text{SGD } 683,790.$

The principal invoice amount of Bond B $= \frac{140}{100} \times 0.9801 \times 500,000 = \text{SGD } 686,070.$

The principal invoice amount of Bond C $= \frac{142}{100} \times 0.9756 \times 500,000 = \text{SGD } 692,676.$

Next, calculate each bond's purchase cost as:

purchase cost = cash dirty price × futures contract size

The purchase cost of Bond A $= \frac{134}{100} \times 500,000 = \text{SGD } 670,000.$

The purchase cost of Bond B $= \frac{135}{100} \times 500,000 = \text{SGD } 675,000.$

The purchase cost of Bond C $= \frac{137}{100} \times 500,000 = \text{SGD } 685,000.$

Finally, calculate each bond's delivery gain/loss as the principal invoice amount – purchase cost.

Delivery gain/loss of Bond A = 683,790 – 670,000 = SGD 13,790.

Delivery gain/loss of Bond B =686,070 – 675,000 = SGD 11,070.

Delivery gain/loss of Bond C =692,676 – 685,000 = SGD 7,676.

The cheapest-to-deliver bond is Bond A, because the gain on delivering it is the highest among the three available bonds.

Compared to Bond A, Bond B has a lower delivery gain on settlement of the futures contract.

Among the three bonds, Bond C has the lowest delivery gain on settlement of the futures contract.

(Module 17.1, LOS 17.a)

Question #62 - 64 of 74

Question ID: 1675234

The decrease in the Singaporean fixed-income portfolio value that results from the 1.5% increase in the interest rate is *closest* to:

A) SGD 4.5 million. ✗

B) SGD 7.2 million. ✗

C) SGD 4.2 million. ✓

Explanation

The change in portfolio value is calculated as:

change in portfolio value = $-MD_T \times$ change in interest rate $\times$ MV

where MD_T is the portfolio target modified duration (the portfolio's duration after accounting for the futures position) and *MV* is the market value of the portfolio.

To derive MD_T, first calculate the target BPV of the portfolio (the BPV of the portfolio after accounting for the futures position) based on the following equation:

$$\text{BPVHR} = \frac{\text{BPV}_\text{T} - \text{BPV}_\text{P}}{\text{BPV}_\text{CTD}} \times \text{CF}$$

where *BPVHR* is the number of future contracts bought or sold, BPV_P is the BPV of the portfolio, BPV_T is the target BPV of the portfolio, BPV_{CTD} is the BPV of the cheapest-to-deliver bond under the futures contract, and *CF* is the conversion factor.

Rearranging the above equation:

$$\text{BPV}_\text{T} = \text{BPV}_\text{P} + \frac{\text{BPVHR} \times \text{BPV}_\text{CTD}}{\text{CF}} = 30,000 + \frac{-22 \times 90}{0.96} = \text{SGD } 27,937.5.$$

Next, calculate MD_T as:

$$\text{MD}_\text{T} = \frac{\text{BPV}_\text{T}}{0.01\% \times \text{MV}}$$

$$\text{MD}_\text{T} = \frac{27,937.5}{0.01\% \times 60,000,000} = 4.65625$$

Thus, the change in portfolio value = $-MD_T \times$ change in interest rate $\times$ MV = –4.65625 × 0.015 × 60,000,000 = –SGD 4,190,625.

The change in portfolio value is miscalculated by using the modified duration of the original portfolio without accounting for the change in modified duration due to the futures position as –5 × 60,000,000 × 0.015 = –SGD 4,500,000.

The change in portfolio value is miscalculated by using the modified duration of the CTD bond underlying the futures contract (rather than using the modified duration of the portfolio after accounting for the futures position) as –8 × 60,000,000 × 0.015 = –SGD 7,200,000.

(Module 17.1, LOS 17.a)

Question #63 - 64 of 74

Question ID: 1675235

The expected net amount paid or received on the first settlement date of the interest rate swap is *closest* to:

A) JPY 4.4 million. ✗

B) JPY 1.5 million. ✗

C) JPY 2.2 million. ✓

Explanation

To decrease the duration of the portfolio, Grid will enter a swap as the party paying the fixed interest rate and receiving the floating interest rate.

To calculate the amount exchanged if Grid used the swap, we need to start by calculating the notional amount of the swap, which is calculated using the following formula:

$$N_S = \frac{MD_T - MD_P}{MD_S} MV_P$$

where N_S is the interest rate swap notional principal, MD_T is the target modified duration of the portfolio, MDP is the current modified duration of the portfolio, MD_s is the modified duration of the swap, and MV_P is the market value of the portfolio:

$$N_S = \frac{4-7}{-2} \times 5.81 \text{ billion} = \text{JPY } 8.715 \text{ billion}$$

To decrease the duration of the portfolio, Grid will have to enter the swap as the fixed rate payer and the receiver of the floating rate.

On the first settlement date:

Amount due on the fixed rate of 1.8%: $0.018 \times = \frac{180}{360} \times 8.715 \text{ billion} = \text{JPY } 78.435 \text{ million}$

Amount to be received: $0.0185 \times = \frac{180}{360} \times 8.715 \text{ billion} = \text{JPY } 80.614 \text{ million}$

Net cash inflow = 80.614 million – 78.435 million = JPY 2.179 million; this is the correct answer.

The cash flows on both legs of the swap are miscalculated using the market value of the portfolio instead of N_S. The amount paid is miscalculated as $0.018 \times = \frac{180}{360} \times 5.81 \text{ billion} = \text{JPY } 52.29 \text{ million}$ and the amount received is miscalculated as $0.0185 \times = \frac{180}{360} \times 5.81 \text{ billion} = \text{JPY } 53.74 \text{ million}$, resulting in net cash inflow = 53.74 million – 52.29 million = JPY 1.45 million.

The swap is assumed to be settled annually instead of semiannually, resulting in amount due = 0.018 × 8.715 billion = JPY 156.87 million, amount to be received = 0.0185 × 8.715 billion = JPY 161.23 million, and net cash inflow = 161.23 million – 156.87 million = JPY 4.36 million.

(Module 17.1, LOS 17.a)

Question #64 - 64 of 74

Question ID: 1675236

The positive difference between the VIX front- month futures price and the VIX spot price *most likely* indicates that the market is in:

A) backwardation and the roll yield is positive. ✗

B) contango and the roll yield is negative. ✓

C) backwardation and the roll yield is negative. ✗

Explanation

Because the front-month futures price is higher than the VIX spot price, the market is in contango.

When the market is in contango, the futures price will be pulled toward the spot rate as the futures contract becomes closes to its maturity, resulting in a negative roll yield.

Because the futures price is higher than the spot price, the market is in contango.

Because the futures price is higher than the spot price, the market is in contango. If the market were in backwardation, the roll yield would have been positive.

(Module 17.4, LOS 17.d)

Question #65 of 74

Question ID: 1675194

A firm has outstanding floating-rate debt on which they pay LIBOR + 200 basis points, and management expects interest rates to increase in the very near future. To create synthetic fixed-rate debt, the *best* strategy for the firm is to enter into a swap in which they:

A) pay fixed and receive floating.

B) pay floating and receive fixed.

C) receive floating and pay floating.

Explanation

To create synthetic fixed-rate debt, the firm should pay fixed and receive floating in a swap. The floating-rate payment they receive in the swap will partially offset the floating rate they pay on their debt. Any portion of the floating rate on the debt that remains (assume 100 bps) will add to the fixed rate they pay on the swap. Their net position on the debt and the swap will be pay fixed + 100 bps = fixed rate.

(Module 17.1, LOS 17.a)

Question #66 of 74

Question ID: 1675206

Simon Feldberg sold treasury futures that have now reached their delivery date. Which of the three eligible bonds below should Feldberg deliver?

	Bond A	Bond B	Bond C
Bond coupon	5.7%	5.75%	5.8%
Bond maturity	20 years	19 years	18 years
Clean price	$144.18	$144.17	$143.96
Accrued interest*	$0	$0	$0
Conversion factor	0.9653	0.9719	0.9782
Futures settlement price		$148.75	
Contract size		$100,000	

Accrued interest* = accrued interest at futures delivery (assumed zero)

A) Bond B.

B) Bond A.

C) Bond C.

Explanation

At the delivery date, the bond that is cheapest to deliver (CTD) will generate the greatest gain or lowest loss on settlement.

$$\text{Profit/(loss) on delivery} = [(\text{Settlement price} \times \text{CF}) + \text{AI}_T] - (\text{CTD clean price} + \text{AI}_T)$$

	Bond A	Bond B	Bond C
Settlement price × CF	$143.59	$144.57	$145.50
Accrued interest	$0	$0	$0
Total	$143.59	$144.57	$145.50
Principal invoice amount **A**	$143,590	$144,570	$145,500
CTD dirty price	$144.18	$144.17	$143.96
Accrued interest	$0	$0	$0
CTD dirty price	$144.18	$144.17	$143.96
Purchase value **B**	$144,180	$144,170	$143,960
Gain/(loss) on delivery (A − B)	($590)	$400	$1,540

Note that this question assumes that accrued interest on the CTD bonds is zero at futures delivery.

> Due to the mark-to-market feature in futures contracts, the short party does not receive the initial futures price at contract expiry. Daily futures settlement prices are used to compute the daily profit or loss on the futures contract with gains being added to the counterparties' margin accounts and losses deducted. Daily profit or loss is calculated as $FP_t - FP_{t-1}$. To avoid double counting profits and losses, the short party will receive the final settlement price × CF at the maturity of the futures contract.

(Module 17.1, LOS 17.a)

Question #67 of 74

Question ID: 1675237

A manager of a $10,000,000 portfolio wants to increase beta from the current value of 0.9 to 1.1. The beta on the futures contract is 1.2, and the futures price is $245,000. Using futures contracts, what strategy would be appropriate?

A) Short 7 contracts.

B) Long 7 contracts.

C) Long 11 contracts.

Explanation

Number of contracts = 6.80 = (1.1 − 0.9) × ($10,000,000) / (1.2 × $245,000), and this rounds up to seven. Because the goal is to increase beta, the manager should go long.

(Module 17.5, LOS 17.e)

Question #68 of 74

Question ID: 1675218

Which of the following statements related to equity swaps is *least accurate*?

A) Equity swaps can be used to gain exposure to equity markets without incurring custody fees.

B) The fixed or floating rate cash outflow that the equity return receiver pays will always be offset by an inflow from the return on the equity index.

C) An investor gaining exposure to equity using swaps will not have voting rights in the underlying equity.

Explanation

The equity return receiver leg of a swap will always pay interest (fixed or floating) and receive any *positive* return on the equity index. If the equity index has a negative return since the last swap settlement date, the equity receiver will have to pay the (negative) return *in addition* to the interest payment.

Equity Swap vs. Physical Stock	
Advantages	**Disadvantages**
• Gain exposure to equity when participation in physical market is restricted	• Equity swaps typically require collateral
• Avoid tax on physical ownership (i.e., stamp duty)	• Swaps are illiquid
• Avoids custody fees on physical ownership	• Swaps do not convey voting rights
• Avoids the cost of monitoring physical positions, which may increase due to corporate actions (e.g., dividend payments, stock splits, buybacks, M&A)	

(Module 17.3, LOS 17.c)

Question #69 of 74

Question ID: 1675243

To synthetically create the risk/return profile of an underlying common equity security:

A) buy the corresponding futures contract and borrow at the risk-free rate.

B) sell short the corresponding futures contract and invest in a T-bill. ✗

C) buy the corresponding futures contract and invest in a T-bill. ✓

Explanation

Futures + cash = security. Therefore, buy the corresponding futures contract and invest in a T-bill.

(Module 17.5, LOS 17.e)

Question #70 of 74

Question ID: 1675242

A portfolio manager knows that a $10 million inflow of cash will be received in a month. The portfolio under management is 70% invested in stock with an average beta of 0.8 and 30% invested in bonds with a duration of 5. The most appropriate stock index futures contract has a price of $233,450 and a beta of 1.1. The most appropriate bond index futures has a duration of 6 and a price of $99,500. How can the manager pre-invest the $10 million in the appropriate proportions?

A) Take a long position in 22 of the stock futures and 25 of the bond futures. ✓

B) Take a short position in 25 of the bond futures and 22 of the stock futures. ✗

C) Take a long position in 25 of the stock futures and 28 of the bond futures. ✗

Explanation

The goal is to create a $7 million equity position with a beta of 0.8 and a $3 million bond position with a duration of 5:

number of stock futures = 21.8 = (0.8 − 0) × ($7,000,000) / (1.1 × $233,450)

number of bond futures = 25.13 = (5 − 0) × ($3,000,000) / (6 × $99,500)

The manager should take a long position in 22 of the stock index futures and 25 of the bond index futures.

(Module 17.5, LOS 17.e)

Question #71 of 74

Question ID: 1675245

A manager of a $2 million fixed-income portfolio with a duration of 3 wants to increase the duration to 4. The manager chooses a swap with a net duration of 2. The manager should become a:

A) receive-floating counterparty in the swap with a notional principal of $1 million. ✗

B) pay-floating counterparty in the swap with a notional principal of $2 million. ✗

C) pay-floating counterparty in the swap with a notional principal of $1 million. ✓

Explanation

To increase duration, the manager should be a pay-floating/receive-fixed counterparty in the swap with a notional principal equal to:

NP = $2,000,000 × (4 − 3) / 2

NP = $1,000,000.

(Module 17.5, LOS 17.e)

Question #72 of 74

Question ID: 1675201

Kevin McMahon believes that Euribor will rise over the next six months and wishes to speculate on this view using three-month Euribor futures. McMahon enters into three futures contracts with 12 months to maturity that have the following characteristics:

Contract size	€1,000,000
12-month futures price	100.350

McMahon closes out his position after six months when the futures price is 99.95 for a contract with six months to expiry. The profit on McMahon's trade is *closest* to:

A) €3,000.

B) €2,250.

C) €1,500.

Explanation

McMahon wishes to speculate that interest rates will rise, and therefore, will initially sell (short) the future. Due to the pricing convention of 100 – annualized forward rate, if forward interest rates rise, the futures price will fall. A price of 100.350 indicates that the forward interest rate was negative at the time of entering the contract.

Note that the term *three-month futures* relates to the interest rate on a three-month deposit starting at a future maturity. The name does not refer to the maturity of the futures contract, which is offered on a quarterly basis and extended for multiple years.

McMahon will close out his futures position after six months by purchasing futures that will mature on the same date as the initial futures he sold. By choosing closeout with futures that have time to maturity as his initial position, the three-month borrowing and lending at maturity offsets perfectly.

Profit/(loss) on contract = (forward rate purchased − forward rate sold) × 90 / 360 × contract size

Sold futures price 100.35 = forward interest rate of −0.35% at contract expiry

Purchased futures price 99.95 = forward interest rate of 0.05% at contract expiry

[(0.0005 − (−0.0035)] × 90 / 360 × €1,000,000 = €1,000 per contract

3 × €1,000 = €3,000

(Module 17.1, LOS 17.a)

Question #73 of 74

Question ID: 1675212

Which of the following statements regarding cross-currency swaps is *least accurate*?

A) If the euro has negative basis against the dollar, the party paying euro interest will pay more than Euribor at each settlement date. ✓

B) In a USD-EUR currency swap, the party receiving dollars at the swap's initiation will pay dollar interest over the life of the swap. ✗

C) Principal payments are exchanged at the beginning of the swap's life, and again at the same exchange rate at the end. ✗

Explanation

In a currency swap, the motivation is to borrow in a cheap currency (low interest cost) and exchange it for the currency desired. The swaps are called cross-currency swaps if both legs' interest payments are based on floating indices (LIBOR, Euribor, etc.).

For example, a European company that requires USD funding may find it cheaper to borrow in euros and then exchange the euros for dollars using a currency swap— relative to borrowing directly in dollars.

The European company will then pay dollar interest and return the dollar principal at the swap's maturity. Note that the currency received on the swap at initiation determines the currency of the interest and principal that will be paid.

The initial principal on the swap is swapped at the initiation date and then again at the end of the swap's tenor. The same exchange rate is used on principal flows at the start and end of the swap's life to eliminate exchange rate risk.

Since 2007, covered interest rate parity (CIRP) has broken down. The result is a difference between interest rates in the currency swap and those used to compute currency forward rates. The dollar lending has attracted higher interest rates than suggested by CIRP. This results in the dollar in the currency swap having positive basis. Typically, basis is quoted on the other currency (foreign) rather than the dollar. Most currencies (but not all) have traded at a negative basis against the dollar since 2007. This means that there is a premium earned for lending dollars in a cross-currency swap. Note that –ve basis on a nondollar currency means the dollar has +ve basis.

For example, EUR − USD currency swap may be quoted at −20 bp. This results in the party paying the euro interest paying Euribor − 20 bp. Remember that the party paying the euro interest will be the party that lent the dollars at the initiation of the swap. Note that +20 bp could have been added to the dollar interest payments; however, in practice, it is typically deducted from the nondollar currency interest payments.

(Module 17.2, LOS 17.b)

Question #74 of 74

Question ID: 1675224

The VIX futures market is currently in backwardation. The VIX is currently at 40, front-month futures are priced at 35, and second-month futures are priced at 30. Assuming the term structure remains constant over the next two months, which of the following trades is *most likely* to result in a position that benefits from roll yield?

A) Buy the second-month future and sell the front-month future. ✓

B) Buy the second-month future and sell the VIX spot.

C) Buy the front-month future and sell the second-month future.

Explanation

The futures price is an expectation of 30-day S&P 500 volatility from futures expiration. The VIX itself is a measure of current 30-day implied forward volatility on the S&P 500.

VIX futures will experience convergence at futures maturity because the future settles against spot VIX at expiration.

The VIX index is normally in contango (futures price > spot VIX), resulting in an upwards sloping term structure. Holding the term structure constant, this means that the futures prices for a specific maturity will decrease overtime. This typically causes negative roll yield. This means purchasers of the VIX futures will make a loss when closing out their position, all else being equal apart from the passage of time.

In times of financial stress, the VIX will spike and can cause backwardation. The backwardation is caused by high levels of expected 30-day volatility, which are expected to decrease overtime. The VIX index is considered mean reverting over time.

In contango markets, the futures price will be pulled down toward the spot, but in a backwardated market, it will be pulled up toward the spot.

Assuming a constant term structure in a backwardated market, the futures price will rise as the basis narrows. The current spread on the second-month future is 40 − 30 = 10. The spread on front-month future is 40 − 35 = 5.

The terms *front-month*, *second-month*, and so on refer to the maturity date of the futures contract.

The strategy in a backwardated market should be to purchase long dated futures and benefit from the convergence to spot over their life (i.e., the decline in basis). This strategy is at risk from sudden changes in the spot volatility of the VIX. The exposure to the spot VIX can be mitigated by selling short dated futures, leaving the investor exposed only to the difference in basis between the two contracts.

The strategy of buying the second-month future and selling the VIX spot is unfeasible as the VIX spot cannot be directly invested in.

The strategy of buying the second-month future and selling the front-month future is profitable in a backwardated market.

The strategy of buying the front-month future and selling the second-month future is profitable in a contango market.

(Module 17.4, LOS 17.d)